ROMHACK
CONFERENCE
TRAINING
CAMP

Andrea Pompili

FROM ZERO TO HERO: The miracle of Plane and Sprite Multiplication



There are only 10 types
of people in the world:
Those who understand binary,
and those who don't

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Load "*",8,1

1985



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1987



I cannot finish this game

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ROMHACK
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All you need is...



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Zzap! Diaries

PROCREAZIONE MENTALE

Tra un'operazione agli occhi e una gita transatlantica per curare l'adozione del suo terzo figlio, il futuro padre Andrew Braybrook salta i corsi pre-natali e si fa venire delle idee di prima mattina...

Parte Terza

di Andrew Braybrook



1. "Bevo perché ne ho bisogno..." Ecco Andrew alla Cinzia Pre-Natale: mestiere per far colpo e conoscere nuovi amici

Venerdì, 13 febbraio

Ancora un altro "Perché non far..." da Gary Liddon, che vorrebbe un campo stellare ruotante invece che uno vettoriale. Questo potrebbe rendere molto più interessante il modo di controllo, dando velocità e rotazione invece di semplici velocità X e Y. Ho attenuto di conseguenza il codice del campo stellare ed ora le stelle invece di ruotare attorno all'astronave si allontanano da essa. Sembra molto strano, come degli scudi antigravitazionali. Alla! Ho invertito i settori e ho ripovato. Successo qualcosa. Chissà cosa succede se rovesci invertito un solo settore? Meglio, non importa neanche quale settore inverti, tranne che cambia la direzione di rotazione. Sfortunatamente il calcolo non è molto accurato poiché le stelle si muovono lentamente a sprizzi verso l'esterno e si raccolgono nell'angolo in alto a sinistra. Serve necessari calcoli più accurati. Ciò nonostante:

... non mi sembra che a scuola mi abbiano mai insegnato a controllare dei campi stellari mobili!

questo dimostra che ST ed io ci ricordiamo un po' di trigonometria di base, anche se non mi sembra che a scuola mi abbiano mai insegnato a controllare dei campi stellari mobili!

Lunedì 16 febbraio

Ho aumentato la precisione del sistema di rotazione stellare e l'ho resa più veloce; ora è tutto più stabile, anche se c'è ancora qualche inaccuratezza. L'ho lasciato girare mentre mangiavo e quando sono ritornato lo schermo era pieno di cerchi e punti e le stelle si erano ruovamente e furtivamente portate verso l'esterno. Frattanto che il gioco stesso tende a scoraggiare lo stare fermi girando su se stessi non vedo alcun problema. Presumo che ora che lo sapete lo proverete tutti.

Per terminare le cose in movimento in un'orbita ovale (ho usato elementi di coordinata leggermente incompatibili) ho ridotto la risoluzione dei movimenti verticali ai pixel alternati. Questo ha il beneficio effetto collaterale di liberare 48 caratteri che erano inservati per quelle posizioni di stelle e mi consente di far girare l'intera giuria di stelle mentre attracco. In precedenza il sistema a griglia mobile usava 32 dei caratteri delle stelle.

Munito dalla voglia di dire a tutti che ho finalmente un Amiga e che è stupendo! L'ho preso di seconda mano da Liddon (comprensivo un computer usato da quest'uomo?) con un'abbondanza collezione di software. Ho usato il Deluxe Paint per fare giochi su schermo di **Morpheus**. È molto utile per fare esperimenti di grafica.

Martedì, 17 febbraio

All'ospedale per una veloce operazione all'occhio: il servizio verrà ripreso appena potrò vedere di nuovo!

Giovedì 26 febbraio

Eccomi di ritorno dopo una settimana di convalescenza con un occhio che sembra come se l'avesse colpito Frank Bruno. Non posso stare davanti alla TV, figuriamoci davanti ad uno schermo di computer. Ho pensato allo scenario del gioco ed ho definito la struttura di gioco, anche se non del tutto. Così, ho deciso di provare a disegnare degli sprite! Nella sequenza di attracco voglio mettere all'estremità dello schermo un'enorme piattaforma di cui si vede però soltanto una parte. Sarà formata da sprite. Ho già disegnato qualcosa e sorvolo sull'Amiga, quindi ho cominciato a disegnare gli sprite sullo sprite editor. È un po' difficile vedere se combaciano così ho scritto un piccolo programma in BASIC per visualizzare come verranno. È andato bene finché non ne avevo più di otto sprite, il mio *sprite editor* no. Considerando i problemi che il CBM BASIC ha a visualizzare anche un solo sprite, non ho molta speranza di riu-

scire a scrivere un [illegible] in BASIC. [illegible] gli sprite come meglio [illegible] tu inserti nel gioco. [illegible] allora, in posizione [illegible] mentre il gioco gira [illegible] do direttamente [illegible] degli sprite sullo schermo. [illegible] piattaforma ho usato in tutto 3 sprite. Non ho fatto grandi arrangiamenti grafici, anche se ci ho messo un po' a mettere a posto un particolare pixel: poi non era altro che un granello di polvere sullo schermo.

...ci ho messo un po' a mettere a posto un particolare pixel: poi non era altro che un granello di polvere sullo schermo...

Questa piattaforma di attracco può ora avere uno schermo com-

Thank you!

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All you need is...

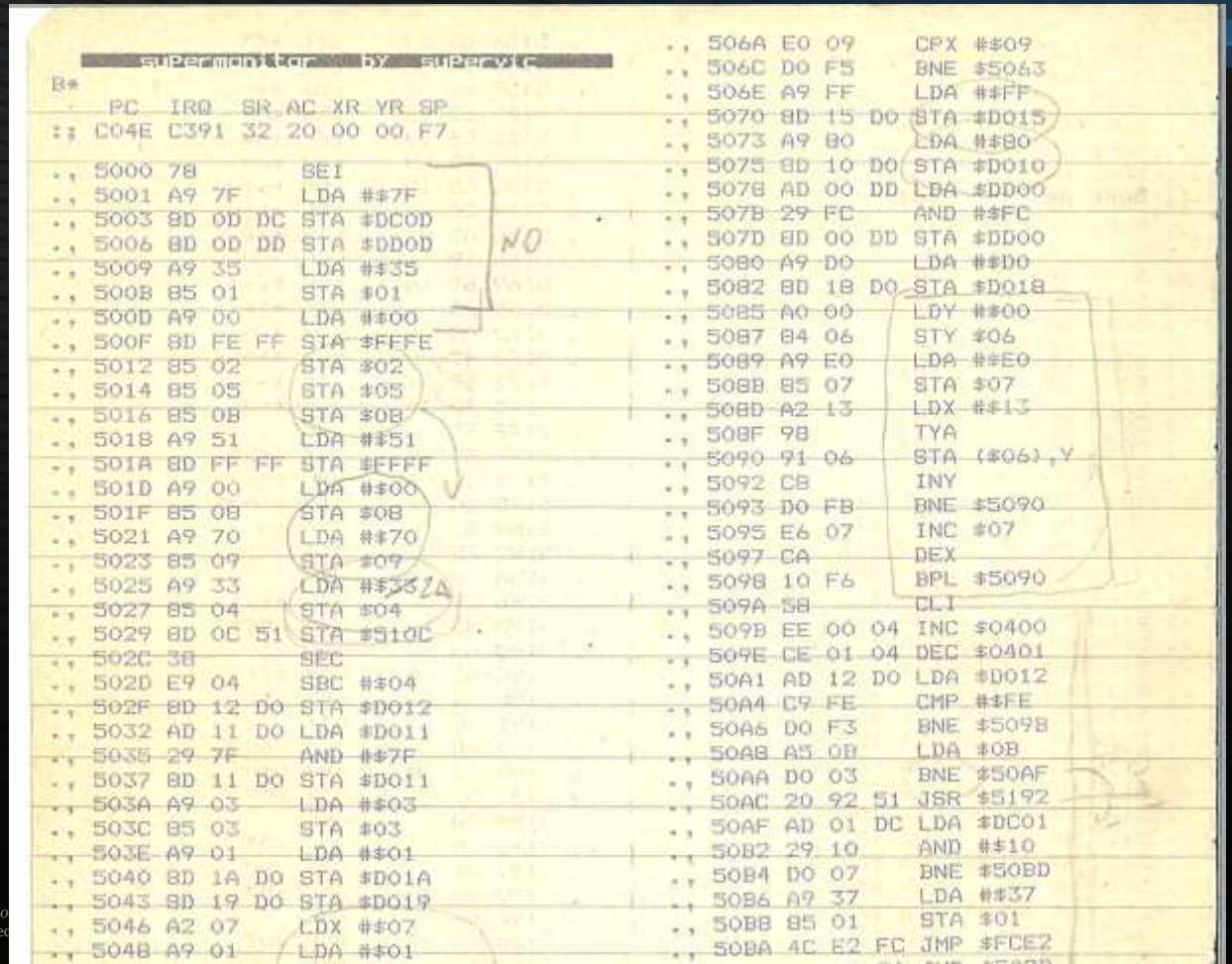


Soft Reset

a.k.a. warm reset or warm start, the motherboard does not lose power during the reset process and so the memory storage remains intact

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```
B*
   PC   IRQ   SR AC XR YR SP
:; C04E C391 32 20 00 00 F7

.; 5000 78       SEI
.; 5001 A9 7F    LDA #$7F
.; 5003 8D 0D DC STA $DC0D
.; 5006 8D 0D DD STA $DD0D
.; 5009 A9 35    LDA #$35
.; 500B 85 01    STA $01
.; 500D A9 00    LDA #$00
.; 500F 8D FE FF STA $FFFE
.; 5012 85 02    STA $02
.; 5014 85 05    STA $05
.; 5016 85 0B    STA $0B
.; 5018 A9 51    LDA #$51
.; 501A 8D FF FF STA $FFFF
.; 501D A9 00    LDA #$00
.; 501F 85 0B    STA $0B
.; 5021 A9 70    LDA #$70
.; 5023 85 09    STA $09
.; 5025 A9 33    LDA #$33
.; 5027 85 04    STA $04
.; 5029 8D 0C 51 STA $510C
.; 502C 38       SEC
.; 502D E9 04    SBC #$04
.; 502F 8D 12 D0 STA $D012
.; 5032 AD 11 D0 LDA $D011
.; 5035 29 7F    AND #$7F
.; 5037 8D 11 D0 STA $D011
.; 503A A9 03    LDA #$03
.; 503C 85 03    STA $03
.; 503E A9 01    LDA #$01
.; 5040 8D 1A D0 STA $D01A
.; 5043 8D 19 D0 STA $D019
.; 5046 A2 07    LDX #$07
.; 5048 A9 01    LDA #$01

.; 506A E0 09    CPX #$09
.; 506C D0 F5    BNE $5063
.; 506E A9 FF    LDA #$FF
.; 5070 8D 15 D0 STA $D015
.; 5073 A9 B0    LDA #$B0
.; 5075 8D 10 D0 STA $D010
.; 5078 AD 00 DD LDA $DD00
.; 507B 29 FC    AND #$FC
.; 507D 8D 00 DD STA $DD00
.; 5080 A9 D0    LDA #$D0
.; 5082 8D 18 D0 STA $D018
.; 5085 A0 00    LDY #$00
.; 5087 84 06    STY $06
.; 5089 A9 E0    LDA #$E0
.; 508B 85 07    STA $07
.; 508D A2 13    LDX #$13
.; 508F 98       TYA
.; 5090 91 06    STA ($06),Y
.; 5092 C8       INY
.; 5093 D0 FB    BNE $5090
.; 5095 E6 07    INC $07
.; 5097 CA       DEX
.; 5098 10 F6    BPL $5090
.; 509A 58       CLI
.; 509B EE 00 04 INC $0400
.; 509E CE 01 04 DEC $0401
.; 50A1 AD 12 D0 LDA $D012
.; 50A4 C9 FE    CMP #$FE
.; 50A6 D0 F3    BNE $509B
.; 50A8 A5 0B    LDA $0B
.; 50AA D0 03    BNE $50AF
.; 50AC 20 92 51 JSR $5192
.; 50AF AD 01 DC LDA $DC01
.; 50B2 29 10    AND #$10
.; 50B4 D0 07    BNE $50BD
.; 50B6 A9 37    LDA #$37
.; 50B8 85 01    STA $01
.; 50BA 4C E2 FC JMP $FCE2
```

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1992



PRESENTAZIONE 90%

Schermate introduttive e finali da capogiro, istruzioni dettagliate, buona suddivisione dei caricamenti e ottima confezione. Basta così?

GRAFICA 94%

Una delle migliori mai apparse sul C64, l'arte italiana si vede!!

SONORO 90%

Buono sotto tutti gli aspetti, ma l'indecente voce digitalizzata all'inizio del gioco potevano risparmiarsela: non potevano mettere una bella scritta? Qui non ci sono mica i sottotitoli alla pag. 777 del televideo!!

APPETIBILITA' 95%

Faaaaaaameeee! E' da tanto che non si vedono più giochi così. Puzzler, puzzler, sempre puzzler... Basta!

LONGEVITA' 94%

Alcuni bug potrebbero pregiudicare questo aspetto del gioco, ma se non vi lasciate scoraggiare, lo giocherete fino a portarlo a termine.

GLOBALE 93%

Un'imperdibile esperienza arcade, interamente realizzata in Italia, che ci fa quasi dimenticare la concorrenza d'oltremanica. Correte subito a comprarlo, o vi mozzo le dita!!

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Miracle #1> The Parallax Scrolling Effect



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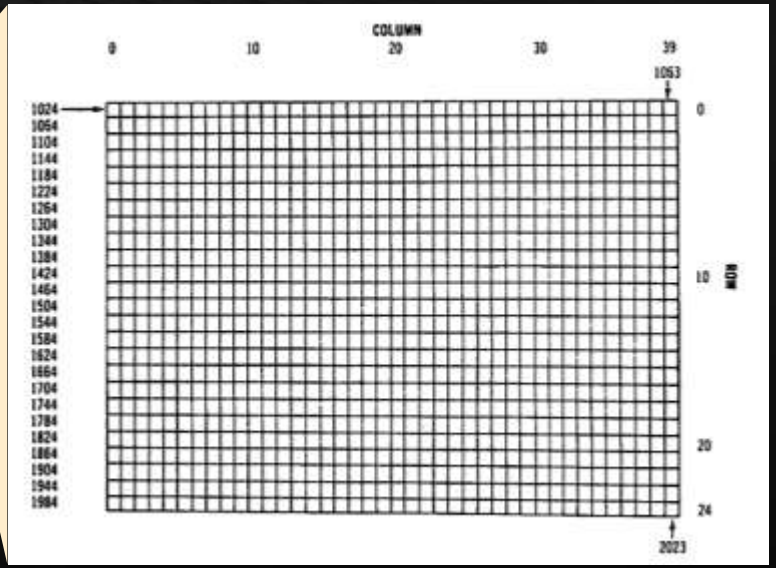
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VIC II Screen Memory Map

Bank 0 16KB	$0000 – $3FFF
Bank 1 16KB	$4000 – $7FFF
Bank 2 16KB	$8000 – $BFFF
Bank 3 16KB	$C000 – $FFFF

Offset 0000	$8000–$83FF
Offset 0001	$8400–$87FF
Offset 0010	$8800–$8BFF
Offset 0011	$8C00–$8FFF
Offset 0100	$9000–$93FF
Offset 0101	$9400–$97FF
Offset 0110	$9800–$8BFF
Offset 0111	$9C00–$9FFF
Offset 1000	$A000–$A3FF
Offset 1001	$A400–$A7FF
Offset 1010	$A800–$ABFF
Offset 1011	$AC00–$AFFF
Offset 1100	$B000–$B3FF
Offset 1101	$B400–$B7FF
Offset 1110	$B800–$BBFF
Offset 1111	$BC00–$BFFF

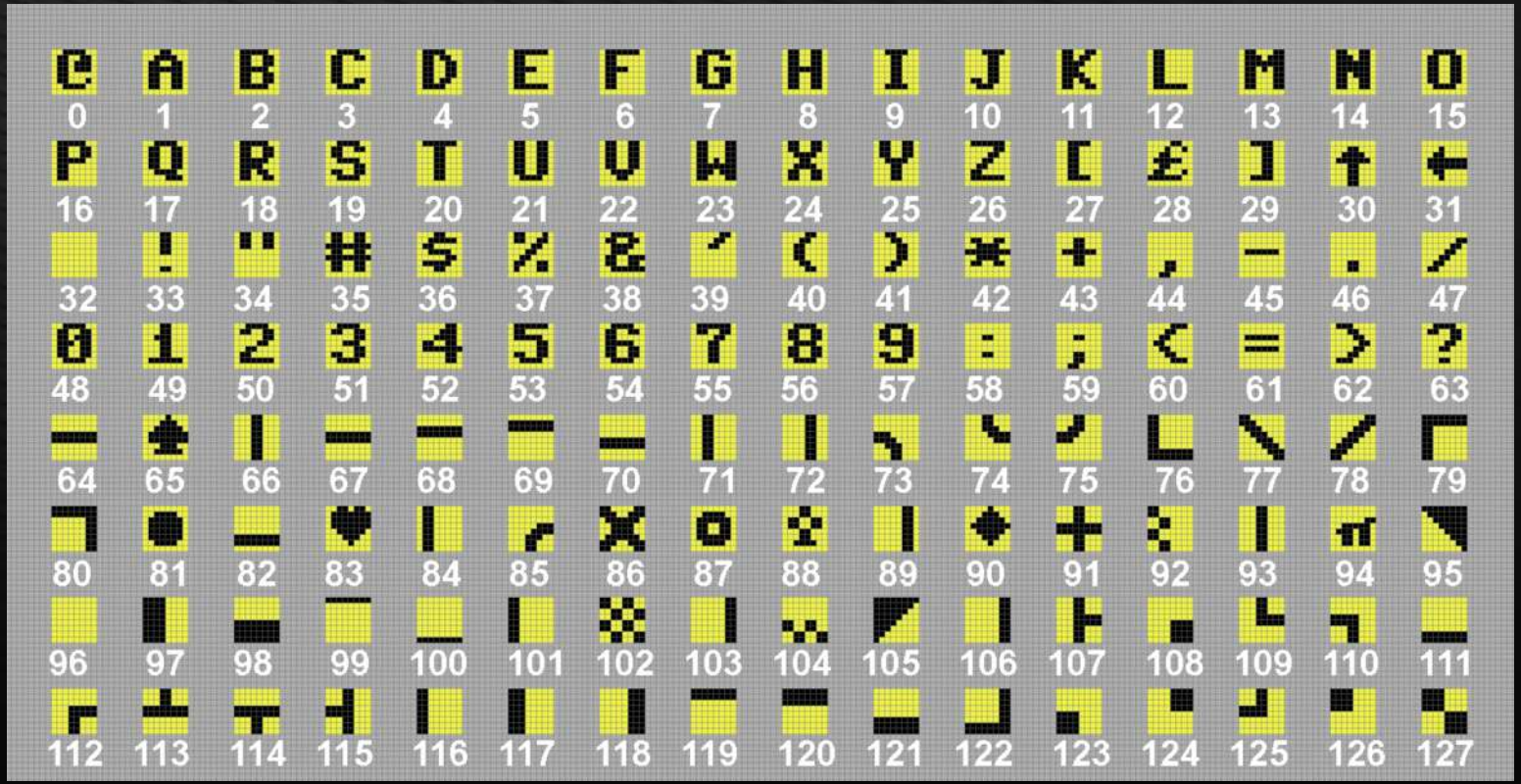


40 x 25 = **1000 ($3E8)**

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PETSCII Character Indexes



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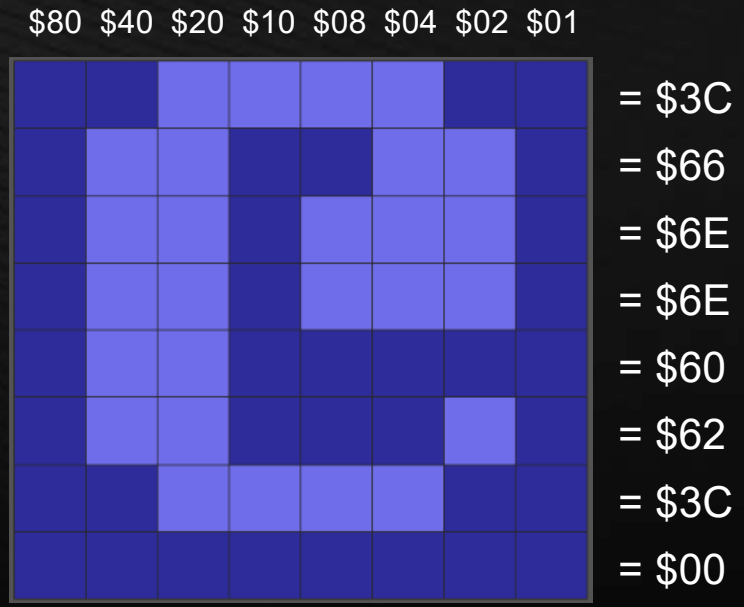
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Custom Characters

Bank 0 16KB	$0000 – $3FFF
Bank 1 16KB	$4000 – $7FFF
Bank 2 16KB	$8000 – $BFFF
Bank 3 16KB	$C000 – $FFFF

Char Map 000	$8000–$87FF
Char Map 001	$8800–$8FFF
Char Map 010	$9000–$97FF
Char Map 011	$9800–$9FFF
Char Map 100	$A000–$A7FF
Char Map 101	$A800–$AFFF
Char Map 110	$B000–$B7FF
Char Map 111	$B800–$BFFF



256 x 8 = **2048 ($800)**

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Multicolor Characters

$80	$40	$20	$10	$08	$04	$02	$01	
0	0	1	0	0	1	0	0	= $24
1	0	1	1	1	0	0	0	= $B8
1	1	1	1	1	0	0	1	= $F9
1	1	1	0	0	1	0	1	= $E5
1	1	1	0	0	1	0	0	= $E4
1	0	0	1	1	0	0	1	= $99
0	1	0	1	0	1	0	0	= $54
0	0	0	1	0	0	0	1	= $11

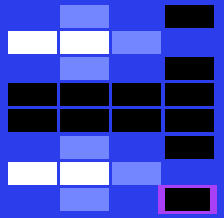
Color Information		
Three shared Colors	$D021 $D022 $D023	Bitmask 00 Bitmask 01 Bitmask 10
individual Color	Color RAM ($D800-$DBFF)	Bitmask 11

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Shortcuts... Character Editor





COLOUR 1

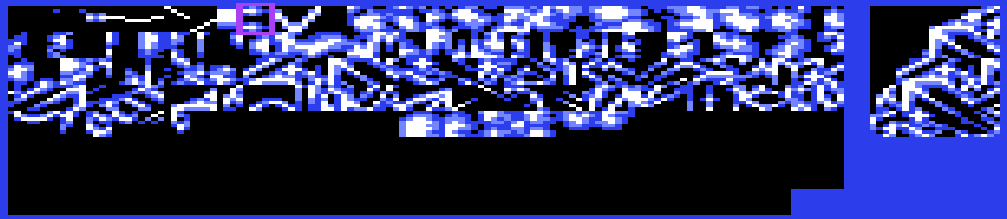
COLOUR 2

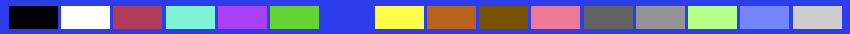
COLOUR 3

BLOCK COL

CHARACTER NO : 009

BLOCK NO : 051





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Shortcuts... Character Editor



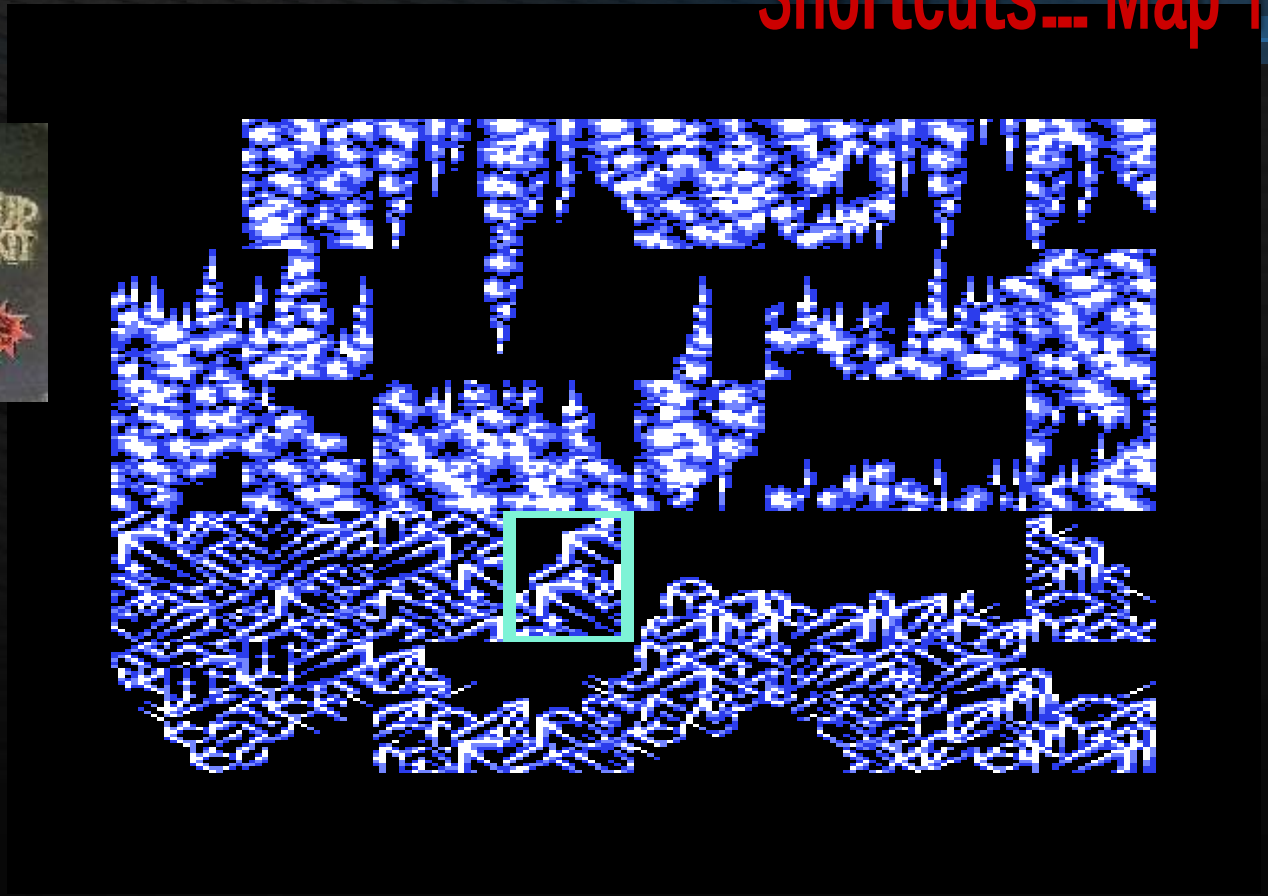


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Shortcuts... Map Tiles (5x5)





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C64 One-direction Scrolling

VIC-II hardware scroll registers (horizontal or vertical)
Allow screen to be offset by up to 7 pixels in each direction

Left scrolling loop:

1. Start setting the horizontal scroll register to **7**
2. Decrement each frame the register by **1** (screen moves 1 pixel to the left)
3. Once below 0, shift the entire RAM screen by 1 byte **(CPU intensive)**
4. Draw the incoming column of tiles
5. Reset scroll register to **7** (screen appeared to move again by 1 pixel)
6. Clear all and repeat

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Flickering

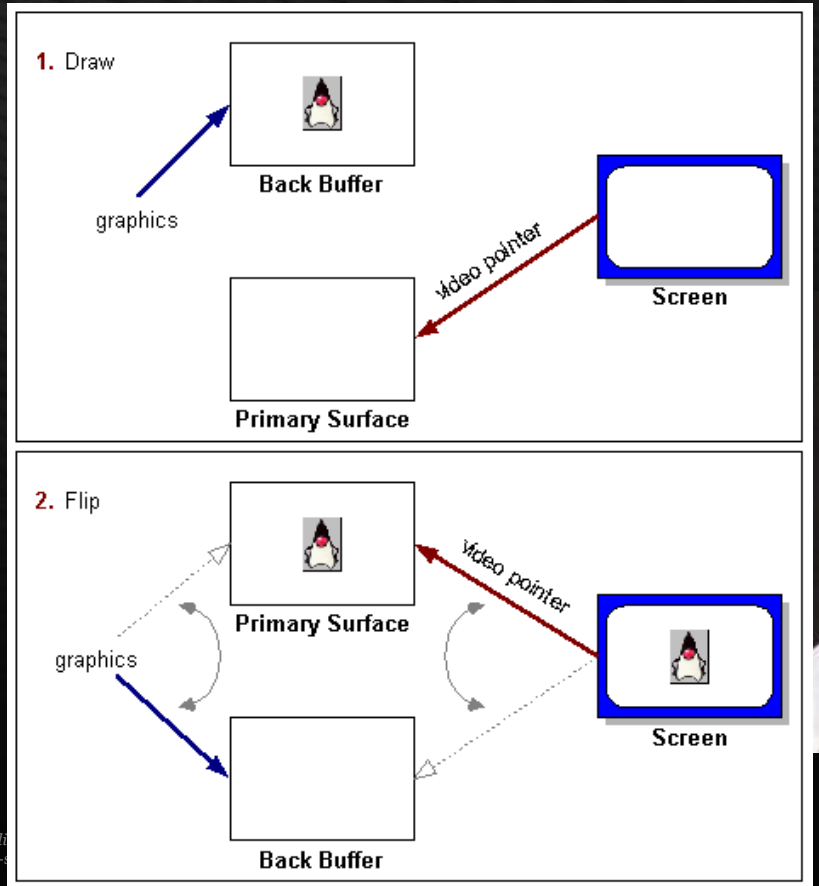


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Don't flicker! Double Buffer!





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Still CPU Intensive...

https://www.c64-wiki.com/wiki/raster_time

Raster time for one horizontal line of graphic (504 pixels including border) equals 63 CPU cycles

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Loop Unrolling

```
// Fine scrolling management section
//
606B C6 02    DEC $02
606D AD 16 D0 LDA $D016         // Set horizontal raster scroll
6070 29 F0    AND #$F0
6072 18       CLC
6073 65 02    ADC $02
6075 8D 16 D0 STA $D016
...

// Move character maps for rough scrolling
6183 A4 02    LDY $02
6185 D0 03    BNE $618A
6187 4C B1 61 JMP $61B1         // #00 > Write the right part of the hidden map
                                //       with the new background content strip
618A A2 00    LDX #$00
618C 88       DEY
618D D0 03    BNE $6192
618F 4C 7E 62 JMP $627E         // #01 > Move map bank #1
6192 88       DEY
6193 D0 03    BNE $6198
6195 4C BF 62 JMP $62BF         // #02 > Move map bank #2
6198 88       DEY
6199 D0 03    BNE $619E
619B 4C 00 63 JMP $6300         // #03 > Move map bank #3
619E 88       DEY
619F D0 03    BNE $61A4
61A1 4C 41 63 JMP $6341         // #04 > Move map bank #4
61A4 88       DEY
61A5 D0 03    BNE $61AA
61A7 4C 82 63 JMP $6382         // #05 > Move map bank #5
61AA 88       DEY
61AB D0 03    BNE $61B0
61AD 4C C3 63 JMP $63C3         // #06 > Move map bank #6
61B0 60       RTS
```

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Loop Unrolling

```
// Move A->B bank #1 (C001:C0A8 -> C400:C4A7)
627E AD 88 02 LDA $0288
6281 D0 1E    BNE $62A1         // check if B map active
6283 BD 01 C0 LDA $C001,X
6286 9D 00 C4 STA $C400,X
6289 BD 2B C0 LDA $C02B,X
628C 9D 2A C4 STA $C42A,X
628F BD 55 C0 LDA $C055,X
6292 9D 54 C4 STA $C454,X
6295 BD 7F C0 LDA $C07F,X
6298 9D 7E C4 STA $C47E,X
629B E8       INX
629C E0 2A    CPX #$2A          // move 42 bytes per segment
629E D0 E3    BNE $6283         // iteration takes 7 CPU cycle (595 saved)
62A0 60       RTS

// Move B->A bank #1 (C401:C4A8 -> C000:C0A7)
62A1 BD 01 C4 LDA $C401,X
62A4 9D 00 C0 STA $C000,X
62A7 BD 2B C4 LDA $C42B,X
62AA 9D 2A C0 STA $C02A,X
62AD BD 55 C4 LDA $C455,X
62B0 9D 54 C0 STA $C054,X
62B3 BD 7F C4 LDA $C47F,X
62B6 9D 7E C0 STA $C07E,X
62B9 E8       INX
62BA E0 2A    CPX #$2A
62BC D0 E3    BNE $62A1
62BE 60       RTS

// Move A->B bank #2 (C0A9:C150 -> C4A8:C54F)
62BF AD 88 02 LDA $0288
62C2 D0 1E    BNE $62E2         // check if B map active
62C4 BD A9 C0 LDA $C0A9,X
62C7 9D A8 C4 STA $C4A8,X
```

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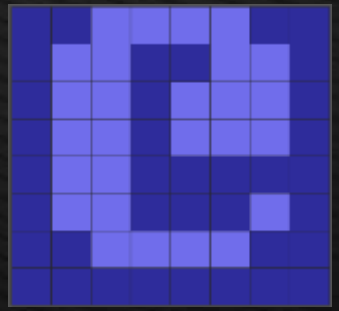
Parallax (1986)



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The Character Rotation trick

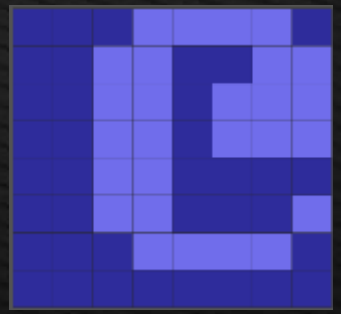


hscrool +0

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The Character Rotation trick



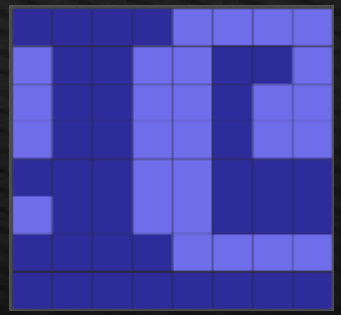
hscrool +1

shift right (ROR)

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The Character Rotation trick



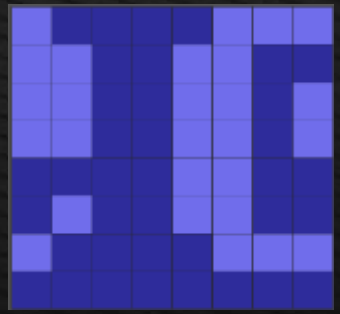
hscrool +2

shift right (ROR)

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The Character Rotation trick



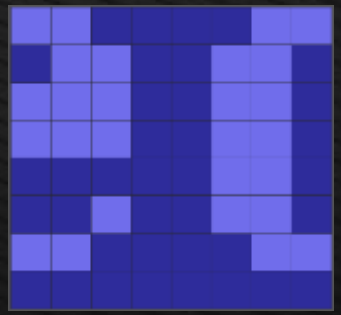
hscrool +3

shift right (ROR)

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The Character Rotation trick



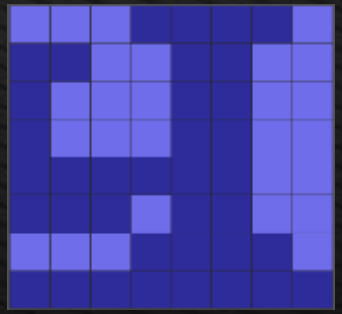
hscrool +4

shift right (ROR)

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The Character Rotation trick



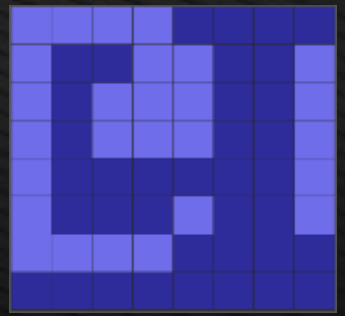
hscrool +5

shift right (ROR)

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The Character Rotation trick



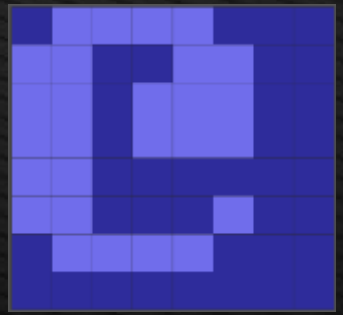
hscrool +6

shift right (ROR)

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The Character Rotation trick



hscrool +7

shift right (ROR)

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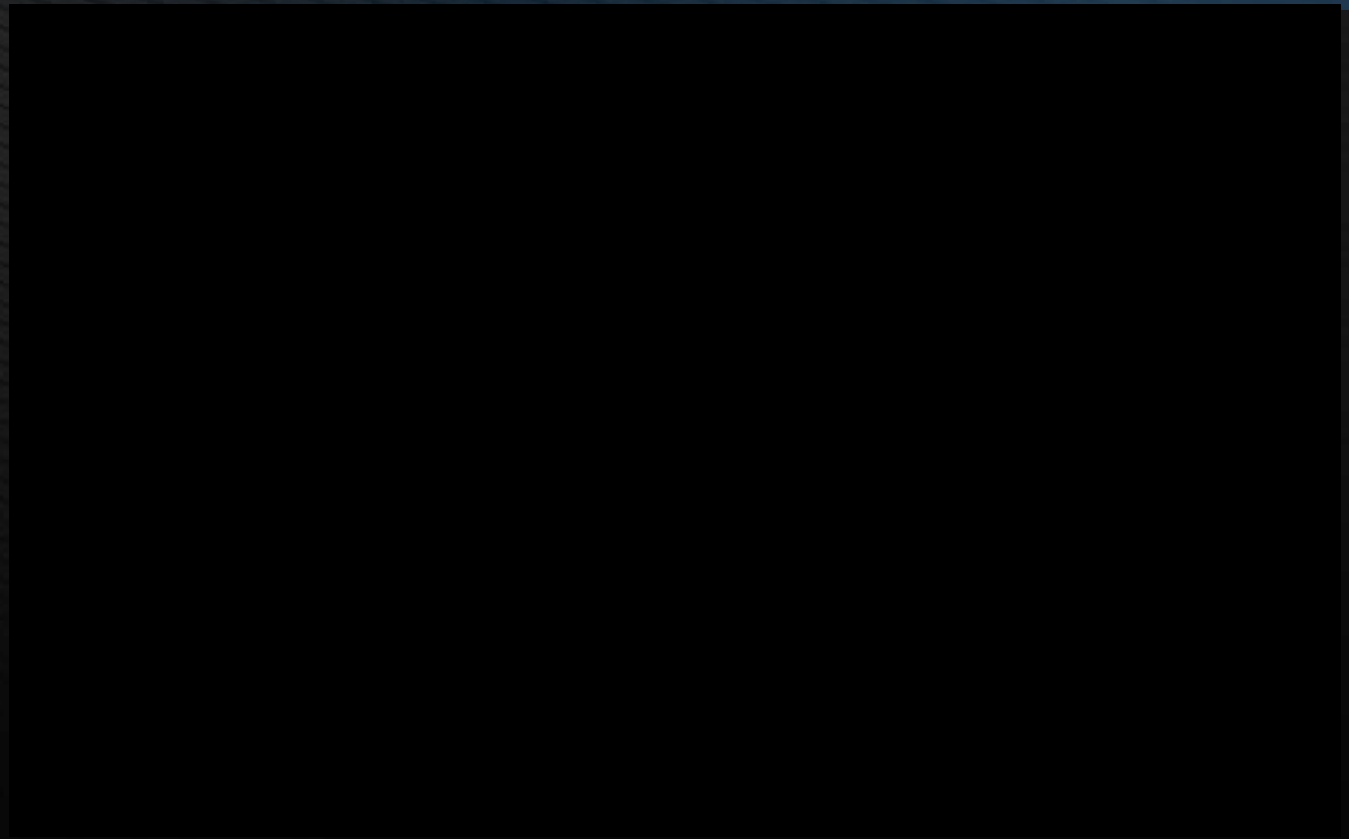
The Character Rotation trick



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Katakis (1988)



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Katakis (1988)



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Phobia (1989)



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Catalypse (1992)



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Boys without Brains (1986)



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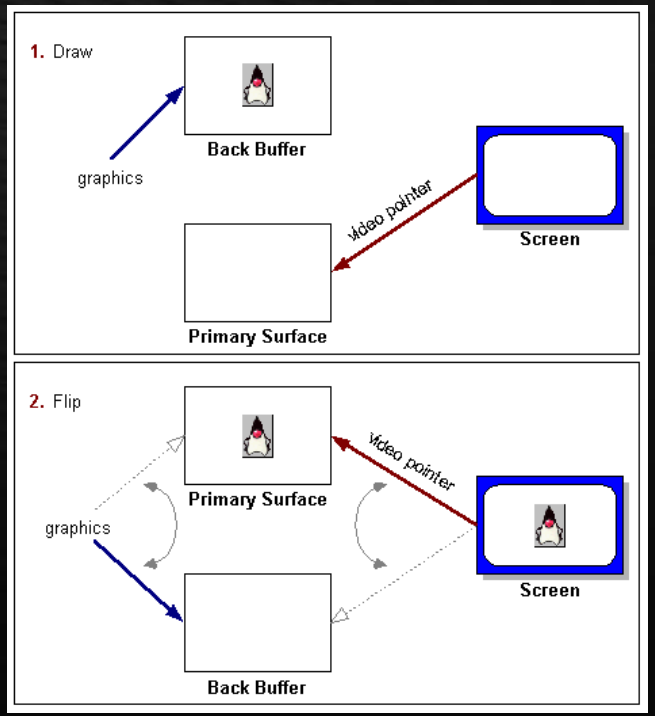
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Multi-Charset Double Buffering

Bank 0	$0000	
16KB		
	$3FFF	
Bank 1	$4000	
16KB		
	$7FFF	
Bank 2	$8000	
16KB		
	$BFFF	
Bank 3	$C000	
16KB		
	$FFFF	

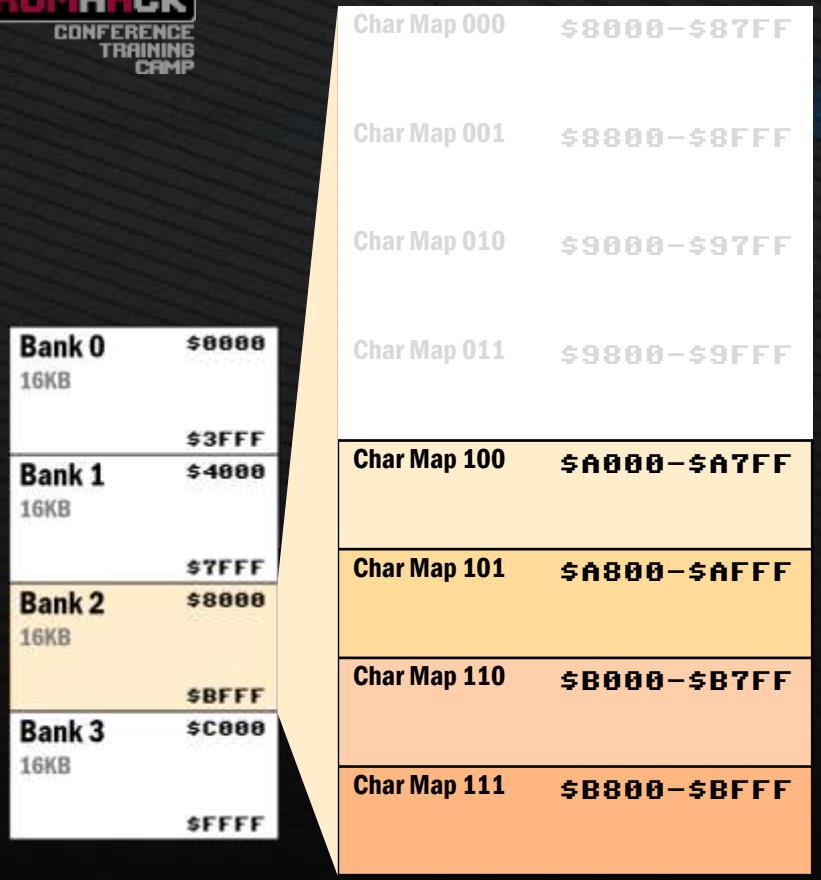
Char Map 000	$8000–$87FF
Char Map 001	$8800–$8FFF
Char Map 010	$9000–$97FF
Char Map 011	$9800–$9FFF
Char Map 100	$A000–$A7FF
Char Map 101	$A800–$AFFF
Char Map 110	$B000–$B7FF
Char Map 111	$B800–$BFFF



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Multi-Charset Parallax



What about rotate multiple charsets?

Charset 1: Background with 0-pixel offset
Charset 2: Background pre-shifted by 2 pixels
Charset 3: Background pre-shifted left by 4 pixels
Charset 4: Background pre-shifted left by 6 pixels

CPU Overhead: Extremely Low

Requires updating one register pointer rather than shifting thousands of bytes of data per frame

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Hawkeye (1988)



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Flimbo's Quest (1990)



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Miracle #2> The Sprite Multiplication



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Sprites

SPRITES

A SPRITE is a special type of user definable character which can be displayed anywhere on the screen. Sprites are maintained directly by the VIC-II chip. And all you have to do is tell a sprite "what to look like," "what color to be," and "where to appear." The VIC-II chip will do the rest! Sprites can be any of the 16 colors available.

Sprites can be used with ANY of the other graphics modes, bit mapped, character, multi-color, etc., and they'll keep their shape in all of them. The sprite carries its own color definition, its own mode (HI-RES or multi-colored), and its own shape.

Up to 8 sprites at a time can be maintained by the VIC-II chip automatically. More sprites can be displayed using RASTER INTERRUPT techniques.

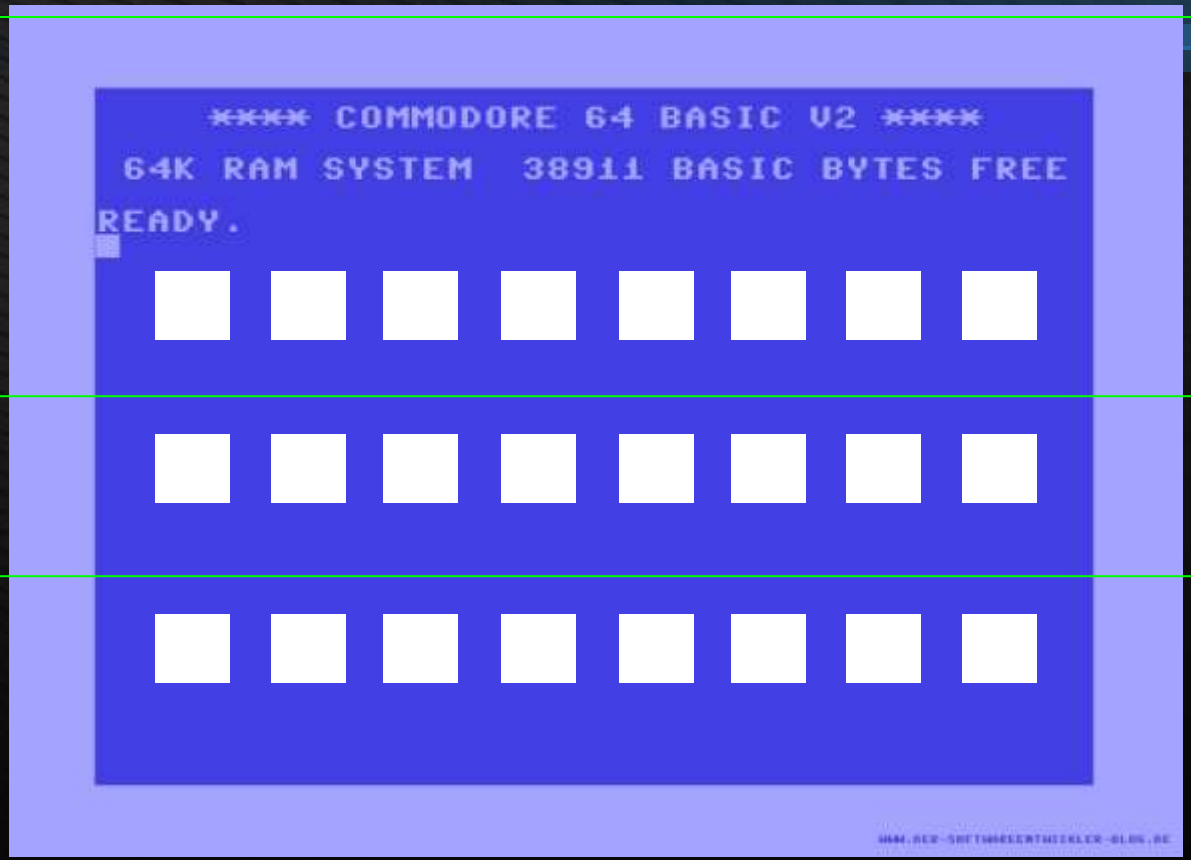
The features of SPRITES include:

1) 24 horizontal dot by 21 vertical dot size.
2) Individual color control for each sprite.
3) Sprite multi-color mode.
4) Magnification (2×) in horizontal, vertical, or both directions.
5) Selectable sprite to background priority.
6) Fixed sprite to sprite priorities.
7) Sprite to sprite collision detection.
8) Sprite to background collision detection.



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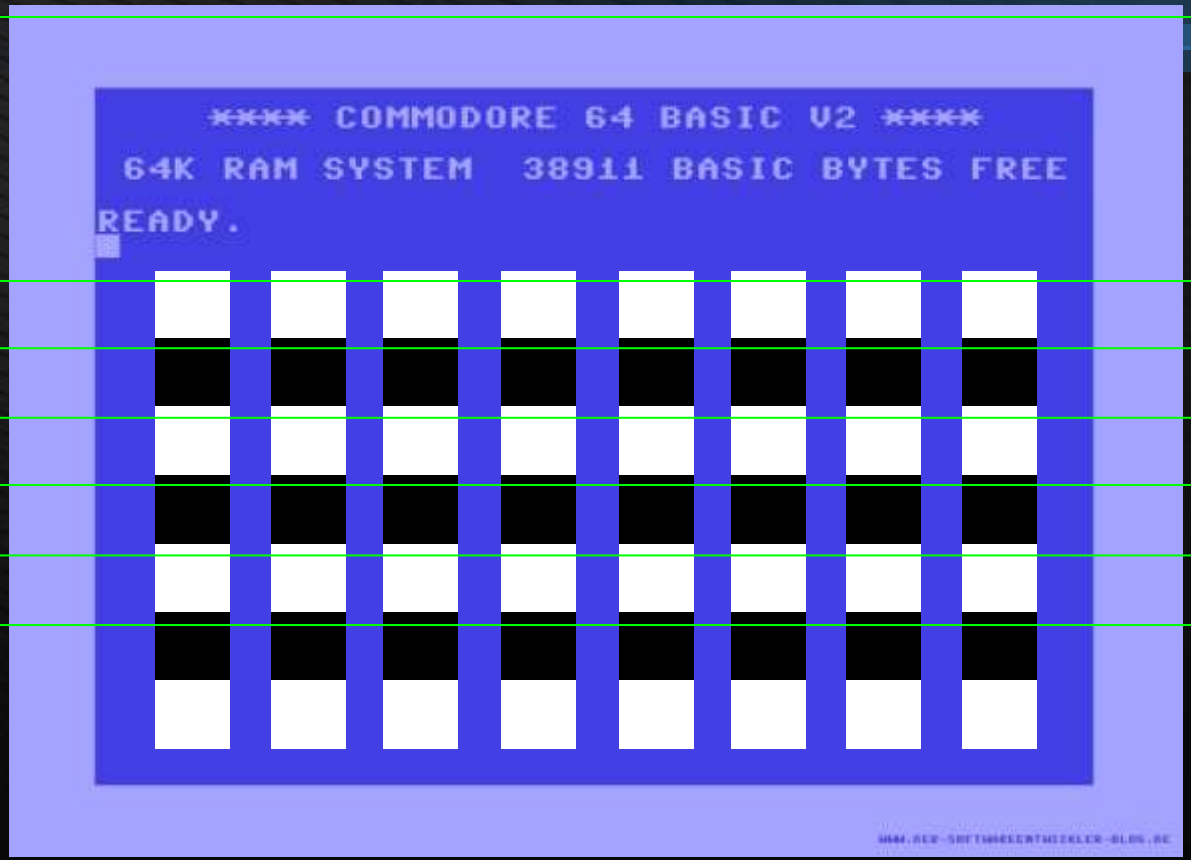
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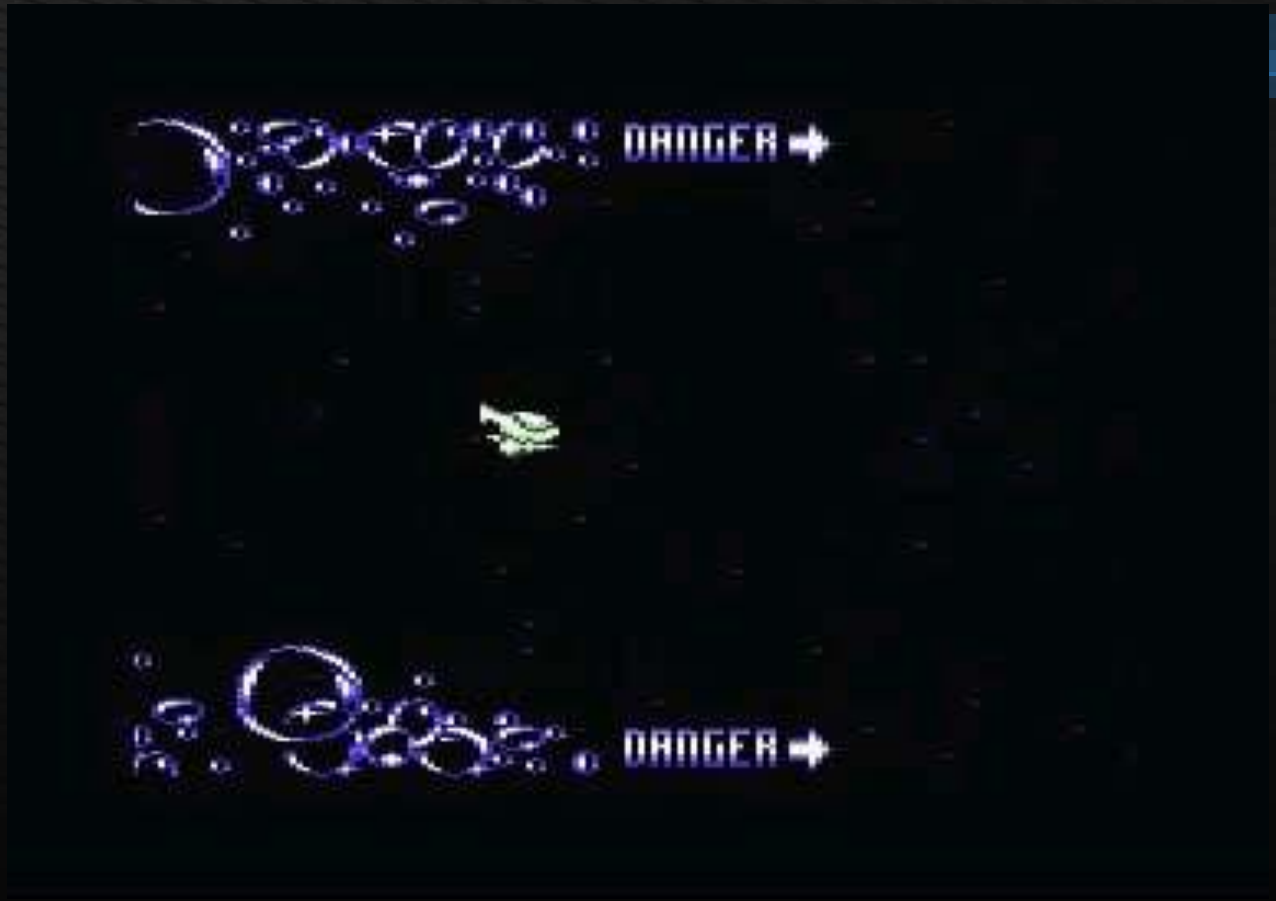
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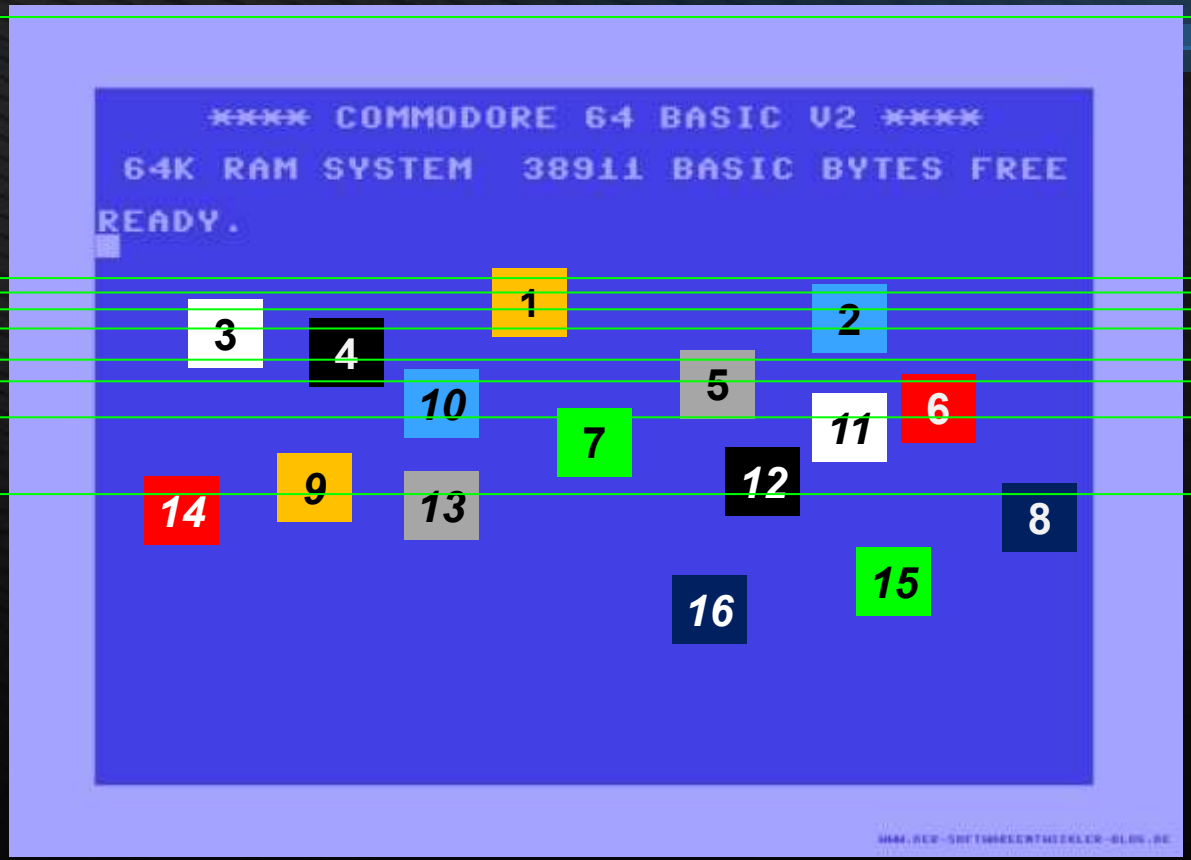
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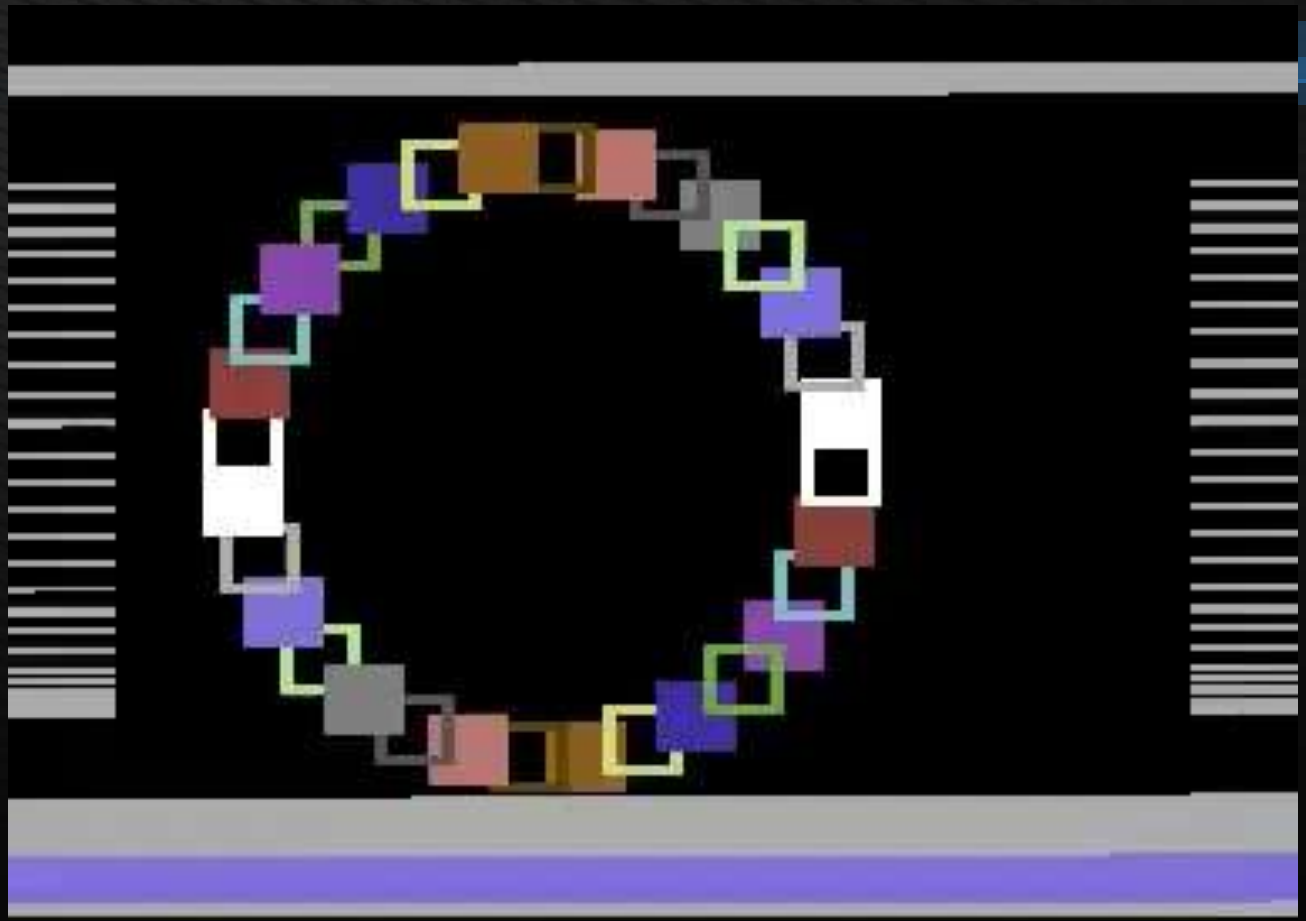
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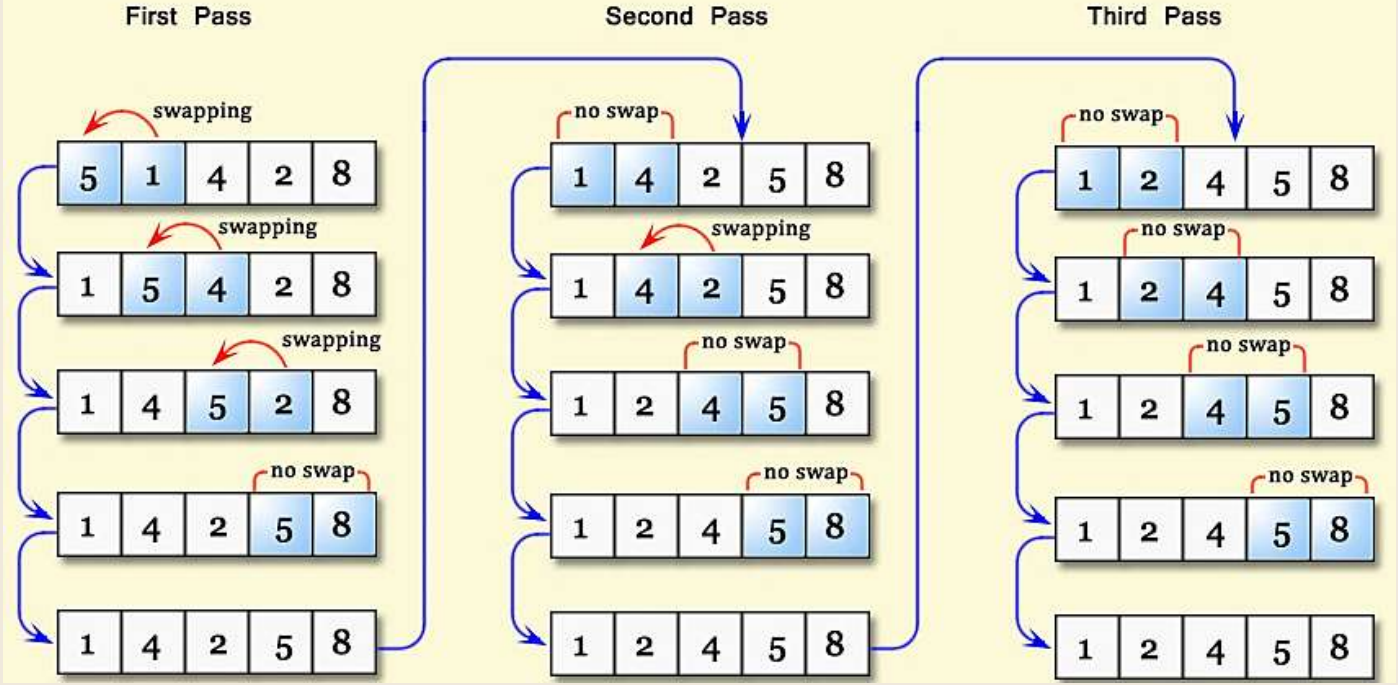
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My Reign for a Sorter => Bubble Sort



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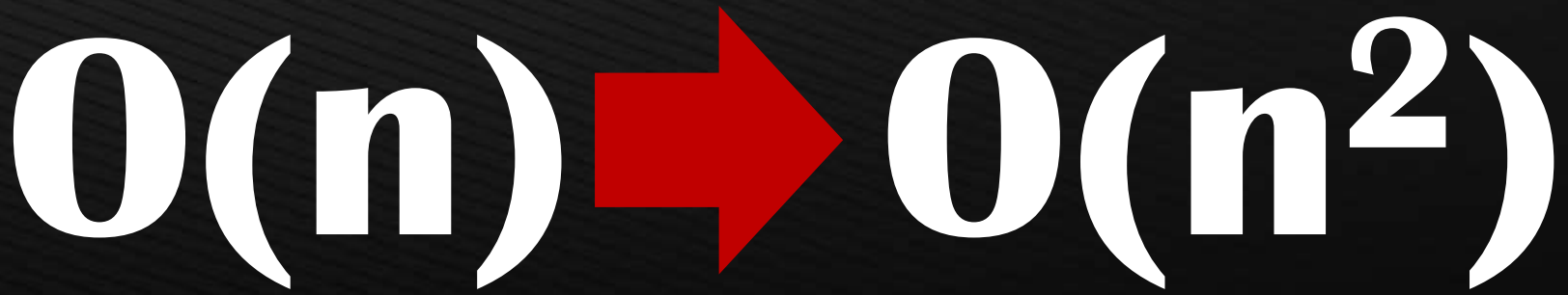
Salamander (1988)



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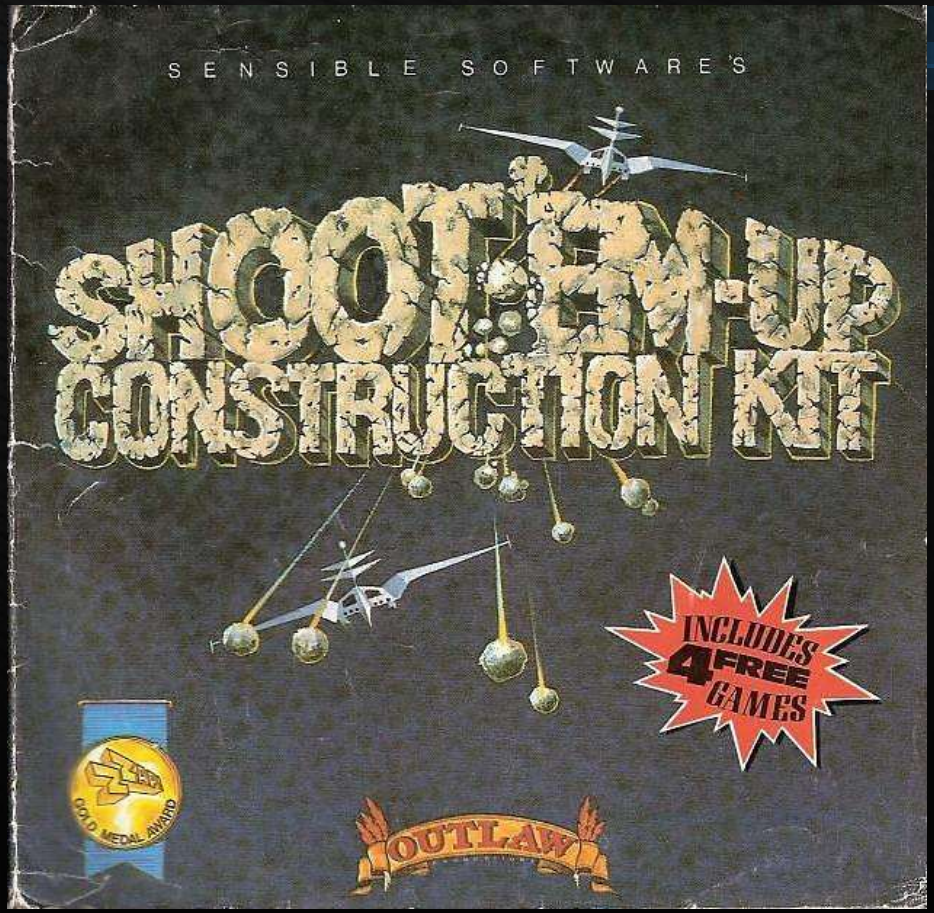
The Bubble Effect...



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SEUCK (1988)



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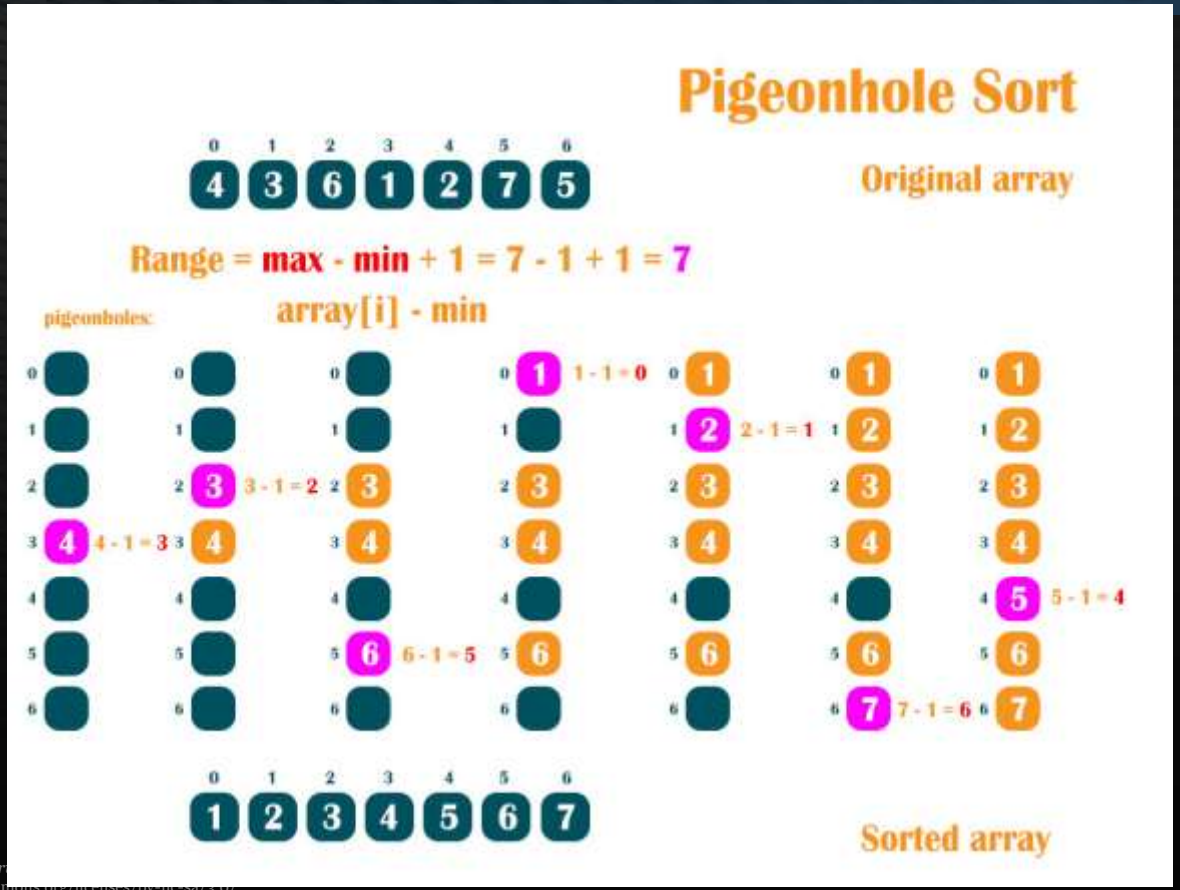
Y-Position is Limited => Pigeonhole Sort



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Y-Position is Limited => Pigeonhole Sort



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Catalypse implementation

```
          LDA   $D012
          BNE   LBL
          LDY   #$00
          STY   $07
          STY   $08
          STY   $09
          STY   $0A
          STY   $0B
          STY   $0C
          STY   $0D
          STY   $0E
          STY   $03
LBL       JMP   NO1
END       STA   $D012
          LDA   $0F
          LDX   $10
          LDY   $11
          RTI
; ROUTINE SPRITE 1
;
NO1       LDX   $03
NO1B      LDA   $07
          CMP   $D012
<ASSEMBLE>
```

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http://creativ

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Catalypse Pigeonhole Sort (Drop in the Bucket)

master mod02.seq

```
// Multiplexer Sorter (Bucket Sort)
// --------------------------------
6D6E A0 00    LDY #$00
6D70 AD 10 A4 LDA $A410         // Y sprite #00
6D73 4A       LSR A             // shift right
6D74 AA       TAX               // load as index

6D75 CA       DEX
6D76 30 0B    BMI $6D83         // go to next sprite
6D78 BD 78 A3 LDA $A378,X       // load bucket from sort buffer
6D7B D0 F8    BNE $6D75         // bucket is not empty, loop back and decrease index

6D7D A9 01    LDA #$01          // get sprite index
6D7F 9D 78 A3 STA $A378,X       // set it to sort buffer
6D82 C8       INY

6D83 AD 11 A4 LDA $A411         // Y sprite #01
6D86 4A       LSR A             // shift right
6D87 AA       TAX               // load as index

6D88 CA       DEX
6D89 30 0B    BMI $6D96         // go to next sprite
6D8B BD 78 A3 LDA $A378,X       // load bucket from sort buffer
6D8E D0 F8    BNE $6D88         // bucket is not empty, loop back and decrease index

6D90 A9 02    LDA #$02          // get sprite index
6D92 9D 78 A3 STA $A378,X       // set it to sort buffer
6D95 C8       INY

6D96 AD 12 A4 LDA $A412         // Y sprite #02
6D99 4A       LSR A             // shift right
6D9A AA       TAX               // load as index

6D9B CA       DEX
6D9C 30 0B    BMI $6DA9         // go to next sprite
```

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Catalypse Pigeonhole Sort (Get and Order)

master mod02.seq

```
6F3E A0 00    LDY #$00
6F40 98       TYA

6F41 C8       INY
6F42 BE 78 A3 LDX $A378,Y       // load current sort buffer
6F45 F0 FA    BEQ $6F41         // loop until a non empty bucket is found
6F47 30 F4    BMI $6F3D         // finished sort buffer... exit
6F49 86 30    STX $30           // save ordered sprite #00 index
6F4B 99 78 A3 STA $A378,Y       // empty the current bucket

6F4E C8       INY
6F4F BE 78 A3 LDX $A378,Y       // load current sort buffer
6F52 F0 FA    BEQ $6F4E         // loop until a non empty bucket is found
6F54 30 E7    BMI $6F3D         // finished sort buffer... exit
6F56 86 31    STX $31           // save ordered sprite #01 index
6F58 99 78 A3 STA $A378,Y       // empty the current bucket

6F5B C8       INY
6F5C BE 78 A3 LDX $A378,Y       // load current sort buffer
6F5F F0 FA    BEQ $6F5B         // loop until a non empty bucket is found
6F61 30 DA    BMI $6F3D         // finished sort buffer... exit
6F63 86 32    STX $32           // save ordered sprite #02 index
6F65 99 78 A3 STA $A378,Y       // empty the current bucket

6F68 C8       INY
6F69 BE 78 A3 LDX $A378,Y       // load current sort buffer
6F6C F0 FA    BEQ $6F68         // loop until a non empty bucket is found
6F6E 30 CD    BMI $6F3D         // finished sort buffer... exit
6F70 86 33    STX $33           // save ordered sprite #03 index
6F72 99 78 A3 STA $A378,Y       // empty the current bucket

6F75 C8       INY
6F76 BE 78 A3 LDX $A378,Y       // load current sort buffer
6F79 F0 FA    BEQ $6F75         // loop until a non empty bucket is found
6F7B 30 C0    BMI $6F3D         // finished sort buffer... exit
```

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Armalyte (1988)

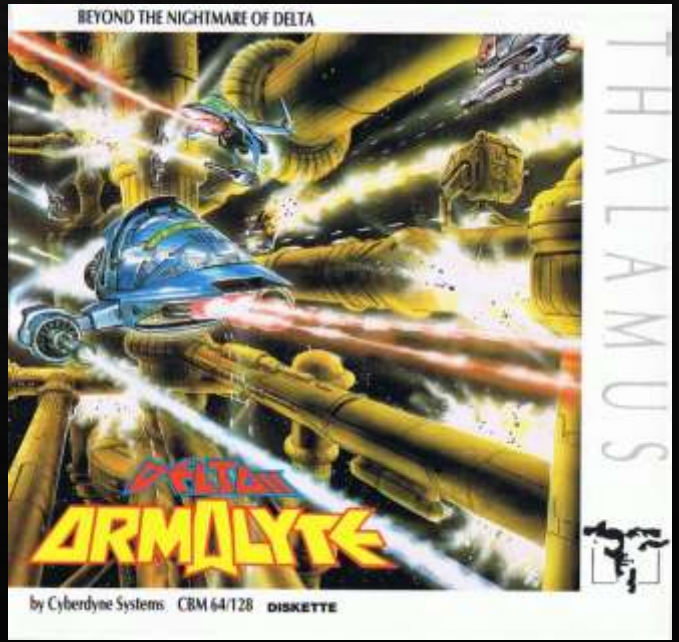
```
NOBJECTS  = 24   ; Number of objects to be sorted
OBJ_YPOS  = $80  ; Y-values of all objects
OBJ_SORT  = $A0  ; Indices of all objects sorted by Y-value
TEMP_SP   = $FF  ; Temporary stack pointer

StackSort:
                ; Save stack pointer
                tsx
                stx TEMP_SP

                ; Phase 1 - Store Y-values in buckets
                .repeat NOBJECTS, i
                  lda OBJ_YPOS + i
                  ;lsr
                  tax
                  txs
                  pla
                  bpl *-1 ; Already value here, try next
                  lda #i
                  pha
                .endrep

                ; Phase 2 - Gather (indices of) sorted values
                ;ldx #(50/2)-1
                ldx #50-1
                txs
                .repeat NOBJECTS, i
                  pla
                  bmi *-1
                  sta OBJ_SORT + i
                  lda #$ff ; Clean up after ourselves
                  pha
                  pla
                .endrep

                ; Restore stack pointer
                ldx TEMP_SP
                txs
                rts
```



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Armalyte (1988)

"The multiplexer was based on work done by Rob Stevens, I just had to make it run faster and combined it with an unrolled version of Gary Liddon's stack bucket sorting routine – as used in Andrew Braybrook's Morpheus, a fab game which I really enjoyed! (We got a naughty alpha copy of Morpheus in the Thalamus offices during the mastering of Hunter's Moon)"

Dan Phillips
March 25th, 2014

http://www.c64.com/gt_display_interview.php?interview=33



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English

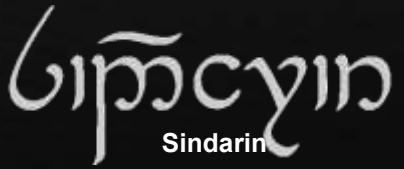
¿Preguntas?
Spanish

مَطَالِب أَيَّة
Arabic

вопросы?
Russian

Domande?
Italian

Ερωτήσεις?
Greek


Sindarin

tupoQghachmey
Klingon

質問
Japanese

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